

Cisco ISR 1111 Router Manual

1. Official Cisco Sources

This manual is based on official Cisco documentation for the Cisco 1000 Series Integrated Services Routers, especially the Cisco 1000 Series ISR Data Sheet, the Cisco 1000 Series ISR support and documentation portal, and the Hardware Installation Guide for the Cisco 1000 Series ISR.

A supporting Cisco router configuration reference was also used to add practical IOS-style configuration steps such as basic router setup, WAN/LAN interface configuration, static routing, command-line access, and recovery behavior. Interface names and port numbers must always be verified on the actual Cisco ISR 1111 using `show ip interface brief`, because interface numbering can differ between router families and variants.

2. Device Description

The Cisco ISR 1111 is a fixed branch router from the Cisco 1000 Series Integrated Services Router family. It runs Cisco IOS XE software and is designed for small and medium branch offices.

Its main role is to connect the internal LAN to the WAN or ISP connection. In a smart office or enterprise branch environment, the ISR 1111 usually sits at the network edge between the ISP/router uplink and the internal Cisco Catalyst switch.

The router can provide:

- WAN connectivity
- Routing between networks
- Static and dynamic routing
- NAT
- DHCP services
- VPN/security services depending on license and configuration
- CLI-based management
- Zero-touch or remote provisioning through Cisco tools

In the project topology, this device should be treated as the **branch edge router**. It connects the smart office network to the internet or WAN and provides the first routing and security layer before traffic reaches the internal switch, access points, AV devices, and endpoint systems.

3. Technical Specifications

The Cisco ISR 1111 belongs to the Cisco C1111 platform. Common variants include models such as C1111-8P and C1111-4P.

The device is a fixed integrated services router intended for branch routing. It is powered by an external AC power supply, depending on the exact model.

The router includes WAN and LAN connectivity. The ISR 1111 platform supports dual Gigabit Ethernet WAN connectivity, usually including one Gigabit Ethernet WAN port and one Gigabit Ethernet WAN/SFP combination port. The LAN side provides integrated Gigabit Ethernet ports, commonly eight ports on the 8-port variant or four ports on the 4-port variant.

Management is available through:

- RJ-45 console port
- USB console
- Cisco IOS XE CLI
- Cisco DNA Center, depending on deployment
- Cisco Plug and Play / zero-touch provisioning, depending on configuration

The base ISR 1111 is not treated as a Wi-Fi access point in this project. Wireless access should be handled by a separate Cisco access point device.

The router can be placed on a desk, shelf, or rack using the proper mounting kit. The C1111-8P chassis is approximately 1.75 x 12.7 x 9.03 inches, including rubber feet.

The platform uses Cisco IOS XE software and requires a supported IOS XE release. The exact supported versions should be checked from Cisco documentation before deployment.

4. Ports, Buttons, and Indicators

The Cisco ISR 1111 includes several physical connection points used for networking, management, power, and recovery.

WAN Ports

The WAN ports connect the router to the ISP, upstream router, modem, firewall, or WAN edge. The device includes a Gigabit Ethernet WAN interface and a Gigabit Ethernet/SFP WAN option depending on the model.

Use the WAN interface for:

- Internet uplink
- ISP handoff
- WAN connection
- Site-to-site network connectivity

LAN Ports

The LAN ports connect the ISR 1111 to internal devices, usually through a downstream switch such as a Cisco Catalyst access switch.

Use the LAN ports for:

- Internal branch network
- Switch uplink
- VLAN routing or routed LAN interfaces
- Local host or lab connectivity

Console Ports

The router includes console access for local administration. Console access is important during first-time setup, password recovery, and troubleshooting when remote access is unavailable.

Available console methods may include:

- RJ-45 console
- USB console

USB Port

The USB port can be used for console functions or storage-related tasks depending on the model and software support. In practical deployment, USB can also support configuration or recovery workflows when supported.

Reset Button

The reset button is used for recovery or factory reset procedures. It should be used carefully because reset or recovery operations may erase or replace the existing configuration.

LEDs

The LEDs provide visual information about:

- Power state
- System status
- Interface link
- Interface activity

During troubleshooting, LED status should be checked before deeper configuration analysis.

5. Installation and Setup Notes

Before installation, confirm that the correct Cisco power supply is available and that the router model matches the intended deployment.

Basic installation steps:

1. Place the router on a stable surface or mount it properly.
2. Connect the Cisco-approved power adapter.
3. Connect the WAN port to the ISP, modem, firewall, or upstream router.
4. Connect the LAN port to the internal switch or local network.
5. Connect to the console port for first-time configuration.
6. Verify interface status using CLI commands.
7. Configure management access, routing, and security.
8. Save the configuration to startup memory.

The router should be installed in a clean, ventilated environment. Since this platform is fanless, proper airflow around the device is important.

6. Basic Initial Configuration

The following commands represent a practical IOS-style initial setup. Interface names are examples only. On the actual ISR 1111, verify the correct names using:

```
show ip interface brief
```

6.1 Enter Configuration Mode

```
enable
configure terminal
```

6.2 Set Hostname

```
hostname ISR1111-Router
```

The hostname makes the router easier to identify in logs, monitoring systems, CLI sessions, and documentation.

6.3 Set Enable Secret

```
enable secret StrongPasswordHere
```

This protects privileged EXEC mode.

6.4 Disable Unwanted DNS Lookup

```
no ip domain-lookup
```

This prevents the router from trying to translate mistyped commands into domain names, which can delay CLI work.

6.5 Save Configuration

```
copy running-config startup-config
```

This step is critical. If the configuration is not saved, it may be lost after reload or power failure.

7. WAN Interface Configuration

The WAN interface connects the router to the ISP or upstream network.

Example:

```
configure terminal
interface GigabitEthernet0/0/0
  description WAN uplink to ISP
  ip address 192.0.2.2 255.255.255.252
  no shutdown
exit
```

After configuration, verify the interface:

```
show ip interface brief
show interfaces GigabitEthernet0/0/0
ping 192.0.2.1
```

Expected result:

- Interface status should be up/up
- IP address should appear correctly
- Ping to the ISP gateway should succeed
- Default route should point toward the WAN gateway

If the interface remains down, check the cable, SFP, upstream device, and whether the interface is administratively shut down.

8. LAN Interface Configuration

The LAN side connects the router to the internal office network or downstream switch.

Example routed LAN interface:

```
configure terminal
interface GigabitEthernet0/1/0
  description LAN connection to internal switch
  ip address 192.168.10.1 255.255.255.0
  no shutdown
exit
```

Verify:

```
show ip interface brief
ping 192.168.10.2
```

Expected result:

- LAN interface should be up/up
- Internal devices should be reachable
- The connected switch should show link activity

If LAN devices cannot communicate, check the cable, interface state, VLAN/subnet design, and IP addressing.

9. Command-Line Access Configuration

Console and remote CLI access should be secured.

9.1 Console Access

```
configure terminal
line console 0
password ConsolePasswordHere
login
exec-timeout 5 30
exit
```

The `exec-timeout` command logs out inactive sessions after the defined time.

9.2 VTY / SSH Access

```
configure terminal
line vty 0 4
password VtyPasswordHere
login
transport input ssh
exec-timeout 5 30
exit
```

For SSH, the router also needs hostname, domain name, user credentials, and RSA keys.

Example:

```
configure terminal
ip domain-name smartoffice.local
username admin privilege 15 secret StrongAdminPassword
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
login local
transport input ssh
exit
```

Verify SSH access from an allowed management device.

10. Static Route Configuration

Static routes define fixed paths for traffic.

A default route is commonly used to send unknown internet traffic toward the ISP gateway.

Example:

```
configure terminal
ip route 0.0.0.0 0.0.0.0 192.0.2.1
end
```

Verify:

```
show ip route
```

Look for:

```
S* 0.0.0.0/0
```

This means a static default route is installed.

Static routes are useful when:

- The branch has a simple topology
- There is one ISP gateway
- The network path does not change often
- The project needs simple and explainable routing logic

Troubleshooting static route problems:

- Confirm next-hop IP is reachable
- Confirm outgoing WAN interface is up
- Confirm route appears in `show ip route`
- Confirm no wrong route points traffic to the wrong gateway

11. Dynamic Routing Configuration

Dynamic routing allows routers to learn and update routes automatically. For a small smart office project, static routing may be enough. However, dynamic routing is useful if the network includes multiple routers or changing paths.

11.1 RIP Example

RIP is simple but older. It may be useful for learning or lab simulation.

```
configure terminal
router rip
version 2
network 192.168.10.0
no auto-summary
end
```

Verify:

```
show ip route
```

RIP-learned routes appear with:

```
R
```

11.2 EIGRP Example

EIGRP is a Cisco routing protocol often used in Cisco environments.

```
configure terminal
router eigrp 100
network 192.168.10.0
network 192.0.2.0
end
```

Verify:

```
show ip route
```

EIGRP-learned routes appear with:

```
D
```

For the project, dynamic routing should only be included if there are multiple routers. If the smart office uses one ISR 1111 router and one switch, static routing is simpler and more suitable.

12. Loopback Interface Configuration

A loopback interface is a logical interface that stays up as long as the router is functioning. It is useful for management, router ID, testing, and stable addressing.

Example:

```
configure terminal
interface Loopback0
 ip address 10.255.255.1 255.255.255.255
exit
```

Verify:

```
show interface loopback 0
ping 10.255.255.1
```

In the project, a loopback can be used as:

- Router management identity
- Stable monitoring target
- Router ID for dynamic routing
- Test interface for troubleshooting

13. Normal Operation

During normal operation:

- Power LED is on
- System LED indicates healthy operation
- WAN interface is up
- LAN interface is up
- Internal hosts receive correct IP addresses
- Router has a valid route to the internet or WAN
- NAT, DHCP, and routing work according to the configuration
- CLI or remote management is accessible to authorized administrators

Useful verification commands:

```
show ip interface brief
show running-config
show startup-config
show ip route
show interfaces
show version
show logging
ping <gateway-ip>
traceroute <destination-ip>
```

14. Troubleshooting

Troubleshooting should move from the physical layer upward:

1. Power
2. Cables
3. Interface status
4. IP addressing
5. Routing
6. NAT/DHCP/security rules
7. Software image

14.1 Device Does Not Power On

Symptom:

- No LEDs
- Router does not boot
- Console shows no output

Possible causes:

- Wrong power adapter
- Loose power connector
- Failed wall outlet
- Faulty power supply
- Hardware failure

Checks:

1. Test the wall outlet with another device.
2. Confirm the Cisco-approved power supply is used.
3. Reseat the DC power connector.
4. Check if LEDs turn on.
5. Try a known-good power supply if available.

Resolution:

If the router remains completely dark with a known-good outlet and power supply, it may require hardware replacement or Cisco TAC escalation.

14.2 WAN Interface Down

Symptom:

- Internet is down
- WAN interface shows down/down or up/down
- LAN traffic may still work internally

Useful commands:

```
show ip interface brief
show interfaces GigabitEthernet0/0/0
show ip route
ping <ISP-gateway>
```

Possible causes:

- Disconnected WAN cable
- Damaged cable
- ISP device down
- Wrong SFP
- Unsupported SFP
- Interface shutdown
- Wrong IP address
- ISP outage

Fix:

```
configure terminal
interface GigabitEthernet0/0/0
no shutdown
exit
```

Then verify physical cabling and test reachability to the ISP gateway.

14.3 LAN Connectivity Problem

Symptom:

- Internal devices cannot reach the router
- Switch uplink does not work
- Users cannot reach local services

Useful commands:

```
show ip interface brief
show interfaces
ping <LAN-host>
show running-config interface <interface-name>
```

Possible causes:

- Bad cable
- Wrong LAN port
- Interface shutdown
- Wrong subnet
- VLAN mismatch
- Switch-side configuration issue

Fix:

```
configure terminal
interface GigabitEthernet0/1/0
no shutdown
exit
```

Confirm that the router LAN IP and client IPs are in the same subnet or that proper routing exists.

14.4 DHCP or IP Addressing Problem

Symptom:

- Clients receive 169.254.x.x addresses
- Clients have no default gateway
- Devices cannot access the network

Possible causes:

- DHCP pool exhausted
- DHCP not configured
- Wrong default gateway
- Wrong subnet mask
- Relay/helper missing
- DHCP excluded-address range too large

Useful commands:

```
show ip dhcp binding
show ip dhcp pool
show running-config | section dhcp
```

Basic DHCP example:

```
configure terminal
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp pool OFFICE-LAN
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit
```

Resolution:

Adjust the DHCP pool, confirm the default gateway, and renew the client IP address.

14.5 Static Route Problem

Symptom:

- Router can reach LAN
- Router can reach WAN gateway
- Internet or remote networks still unreachable

Useful commands:


```
show ip route
ping <next-hop>
traceroute 8.8.8.8
```

Possible causes:

- Missing default route
- Wrong next-hop IP
- WAN gateway unreachable
- Route points to wrong interface
- Upstream router lacks return route

Fix example:

```
configure terminal
ip route 0.0.0.0 0.0.0.0 <ISP-gateway-IP>
end
```

14.6 CLI or SSH Access Problem

Symptom:

- Router forwards traffic
- Admin cannot access CLI remotely
- SSH connection fails

Possible causes:

- SSH not enabled
- RSA keys missing
- Wrong username/password
- VTY line misconfigured
- ACL blocking admin subnet
- Management IP unreachable

Useful commands from console:

```
show running-config | section line vty
show ip ssh
show users
show access-lists
show ip interface brief
```

Fix:

```
configure terminal
ip domain-name smartoffice.local
username admin privilege 15 secret StrongAdminPassword
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
login local
transport input ssh
end
```

14.7 Software or Boot Issue

Symptom:

- Router does not boot normally
- Router reloads repeatedly
- IOS XE image appears corrupted
- Console shows boot errors

Possible causes:

- Corrupt IOS XE image
- Unsupported image
- Flash issue
- Interrupted upgrade
- Wrong boot variable

Useful commands:

```
show version
show flash:
show boot
```

Resolution:

Load a supported IOS XE image, verify image integrity, and confirm the boot variable.

14.8 Configuration Lost After Reload

Symptom:

- Router returns with default or old configuration after reload
- New changes disappear

Possible causes:

- Running configuration was not saved
- Startup configuration was erased
- Wrong configuration register after recovery
- Factory reset was performed

Fix:

```
copy running-config startup-config
```

Verify:

```
show startup-config
show version
```

15. Recovery and Reset

The reset button can be used for recovery, but it should be handled carefully.

Recovery may be used when:

- Admin access is lost
- Startup configuration is corrupted
- Router needs to load a known configuration
- Router requires factory reset

General recovery concept:

- During boot or recovery, the router may load a default image or a specific configuration file if one is available.
- When IOS is already running, pressing the reset button for a defined period may trigger the router to search for a recovery configuration file.
- If a valid configuration file exists in USB flash or internal flash, the router may copy it into startup configuration and reload.
- If no valid configuration file exists, the reset action may have no useful effect or may factory reset depending on platform behavior.

Important warning:

Before using reset or recovery, back up the current configuration whenever possible.

Backup command:

```
copy running-config startup-config
copy running-config flash:backup-config.cfg
```

Factory reset or recovery should be documented because it can erase or replace the router configuration.

16. Zero-Touch Deployment / USB Configuration Concept

Zero-touch deployment allows a router with no startup configuration to load a prepared configuration file automatically, usually from USB or a provisioning system.

General requirements:

- Router has no startup configuration
- USB flash drive is inserted, if USB deployment is used
- Valid `.cfg` configuration file exists
- Configuration file is plain text

- Only the intended configuration file should be present to avoid loading the wrong file

This is useful for:

- Fast branch deployment
- Replacing failed routers
- Standardizing smart office network setups
- Reducing manual configuration errors

In the smart office project, this can be described as a practical deployment method for quickly installing the ISR 1111 with a prebuilt configuration.

17. Maintenance and Best Practices

Recommended practices:

- Keep IOS XE on a supported release.
- Back up the running configuration after changes.
- Keep a copy of the startup configuration.
- Label WAN and LAN cables.
- Document interface usage.
- Keep the router in a ventilated area.
- Avoid blocking vents.
- Use supported SFPs and cables.
- Restrict management access to trusted admin networks.
- Use SSH instead of Telnet.
- Use strong local credentials.
- Monitor logs and interface errors.
- Save configuration before reload.
- Document every recovery or reset action.

Useful maintenance commands:

```
show version
show logging
show interfaces
show ip interface brief
show running-config
show startup-config
show flash:
copy running-config startup-config
```

18. Quick Reference

Power Check

```
Check power adapter
Check outlet
Check power LED
Check console boot output
```

Interface Check

```
show ip interface brief
show interfaces
```

Routing Check

```
show ip route
ping <next-hop>
traceroute <destination>
```

Configuration Check

```
show running-config
show startup-config
```

Save Configuration

```
copy running-config startup-config
```

WAN Setup

```
interface GigabitEthernet0/0/0
ip address <WAN-IP> <MASK>
no shutdown
```

LAN Setup

```
interface GigabitEthernet0/1/0
ip address <LAN-IP> <MASK>
no shutdown
```

Static Default Route

```
ip route 0.0.0.0 0.0.0.0 <ISP-GATEWAY>
```

SSH Setup

```
ip domain-name smartoffice.local
username admin privilege 15 secret <PASSWORD>
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
login local
transport input ssh
```

DHCP Check

```
show ip dhcp binding
show ip dhcp pool
```

Recovery

```
Use console access first.
Back up configuration if possible.
Use reset/recovery only when required.
Document the recovery action.
```

19. Final Notes

For the smart office project, the Cisco ISR 1111 should be documented as the main routing and WAN-edge device. The most important practical data is not only the hardware description, but also the operational commands, verification commands, troubleshooting flows, and recovery procedures.

The best project-ready structure is:

1. Device identity
2. Network role
3. Ports and interfaces
4. Setup steps
5. Verification commands
6. Troubleshooting symptoms

7. Causes
8. Fix actions
9. Recovery methods
10. Quick reference commands